Predictive modelling using genome-wide DNA methylation data

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Background

- In search of predicative methylation signatures for complex diseases, recent studies have identified valuable DNA methylation-based biomarkers. Most previous studies have built prediction models based on single CpGs, without considering the methylation status of neighboring CpG sites.
- However, compared to single CpGs, differentially methylated regions (DMRs) give higher confidence and likelihood of biological importance.
- We tested the hypothesis that the accuracy of Elastic Net prediction models can be improved by using predictors based on co-methylated regions (clusters).

Method

- **Datasets:** We used two publicly available DNA methylation datasets, the AddNeuroMed dataset (GEO accession: GSE144858) with 171 samples (88 CN, 83 AD), and the AIBL dataset (GEO accession: GSE153712) with 491 samples (356 CN, 135 AD)
- **Pre-processing of DNA methylation datasets:** as previously described in Zhang et al. (2020) (PMID: 33257653). Co-methylated regions were identified using CoMethAllRegions() function in coMethDMR R package.
- **Summary of clusters:** We compared several different ways to represent each co-methylated cluster: by using median methylation level, mean methylation level, most variable CpG, CpG with max. methylation level, first principal component, or all CpGs
- **Internal Validation:** Within each dataset, we performed five-fold cross-validation (i.e., 4 folds of the samples were randomly selected to determine tuning parameters, the resulting model was then tested on the remaining 1 fold of the samples). This process was then repeated 10 times to generate 50 estimated AUCs and their average.
- **External Validation:** Prediction model was built using all samples from one dataset and then tested on the remaining dataset.

Abbreviations: **CN** = Cognitive Normal, **AD** = Alzheimer's dementia

Results: cluster-based models

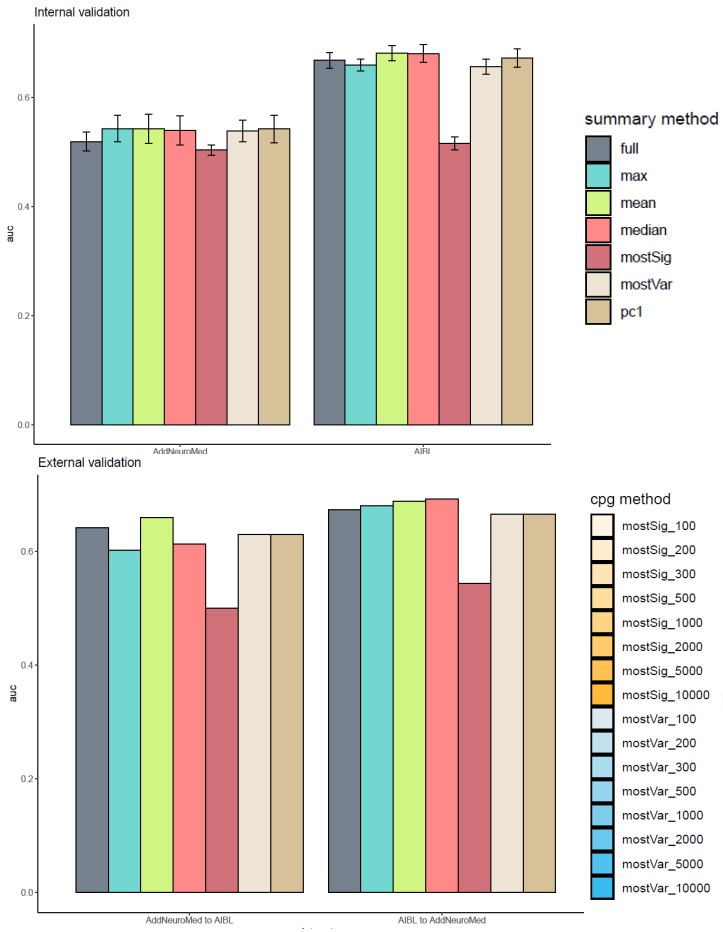

Fig 1 Performance of Elastic Net prediction models using only co-methylated clusters as predictors

Results: CpG-based models

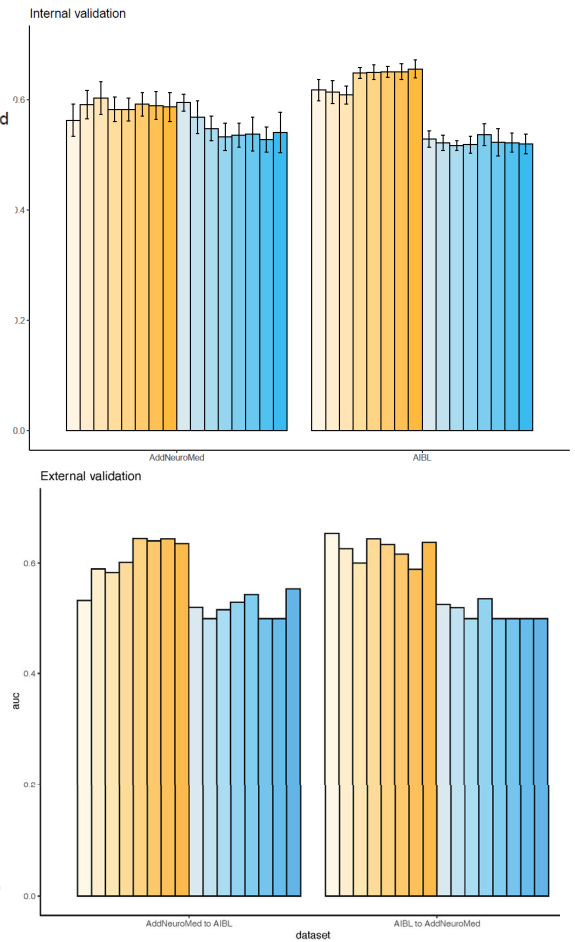

Fig 2 Performance of Elastic Net prediction models using most significant or most variable CpGs.

Conclusions

- The prediction performance for cluster-based Elastic Net models performed similarly as models using only the CpGs as predictors.
- Among different summarization methods for clusters, models that used the most significant CpG to represent each cluster performed the worst. Other summary methods performed similarly
- For models that used only the CpGs, the models that used the most significant CpGs performed uniformed better than those using most variable CpGs.

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Nothing to disclose